

Federated Learning-Based Diabetic Retinopathy Detection Using CNN with SCAFFOLD, Personalization, and Differential Privacy

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Abstract

Diabetic retinopathy (DR) is a leading cause of preventable blindness worldwide. Early detection is critical, but most deep learning systems require centralizing patient images, which conflicts with HIPAA and GDPR. This paper proposes a federated learning (FL) system for five-class DR grading using the Indian Retinal Image Dataset (IRDRD). The system combines three key additions: (1) SCAFFOLD to fix client drift on Non-IID hospital data, (2) local fine-tuning for per-hospital personalization, and (3) DP-SGD for formal differential privacy guarantees. A direct comparison of FedAvg versus SCAFFOLD on the same IRDRD partitions is conducted. Expected results show faster convergence, higher accuracy, and full privacy compliance with SCAFFOLD.

Keywords: Diabetic Retinopathy, Federated Learning, SCAFFOLD, FedAvg, Differential Privacy, Personalized FL, CNN, IRDRD, Non-IID

. INTRODUCTION

A. Background of Diabetic Retinopathy

Diabetes mellitus is a long-term condition affecting over 500 million people globally. One of its most serious complications is diabetic retinopathy (DR), which damages blood vessels in the retina. If untreated, DR leads to permanent blindness. It is classified into five stages: No DR (Grade 0), Mild NPDR (Grade 1), Moderate NPDR (Grade 2), Severe NPDR (Grade 3), and Proliferative DR (Grade 4) [10].

About 103 million people currently have DR, and this number may reach 700 million by 2045 [3]. India has one of the world's largest diabetic populations, yet access to eye doctors and retinal imaging equipment in rural areas is very limited.

B. Importance of Early Detection

DR has no clear symptoms in early stages, so patients often notice vision loss only when it is too late for effective treatment. Catching DR early allows management through laser therapy, anti-VEGF injections, and lifestyle changes. Manual screening by ophthalmologists is slow and cannot reach the full at-risk population. CNNs have shown diagnostic accuracy comparable to expert clinicians [1], making automated screening a practical solution. However, these models require large centralized datasets, which conflicts with patient privacy regulations.

C. Role of Federated Learning

Federated learning (FL) allows multiple hospitals to jointly train a model without sharing raw patient data [5]. Each client trains a local model on its own data and sends only weight updates to a central server. The server aggregates these updates into an improved global model, which is sent back to all clients. This setup satisfies HIPAA and GDPR requirements because no patient images ever leave the originating hospital [8].

D. Problem Motivation

The most common FL algorithm, FedAvg, assumes data is distributed identically across clients (IID). Real hospital data is not IID — different hospitals have different patient groups, equipment, and practices. This heterogeneity causes *client drift*, where local models diverge from the global optimum, slowing convergence [5].

Additionally, most FL-DR papers have not tested on the IRDRD dataset (Indian patients), and only one reviewed paper (P12) applies formal differential privacy. No paper combines SCAFFOLD, personalization, and differential privacy in a single system. This research fills all three gaps.

. LITERATURE REVIEW

A. Overview of Reviewed Works

Fifteen papers on FL for DR detection were reviewed, spanning 2020 to 2025. Early works (P6, P7) showed FL is feasible for DR. P7 achieved 97.87% accuracy using FedSGD; P6 combined transfer learning with FedAvg and reached 90.07%.

Recent papers pushed further. P10 modified FedAvg's loss function to reach 98.6% across five clients. P14 used FedProx with MobileNet and reached 98.36% with 20 clients. P2 added homomorphic encryption to reach 98.6% with reduced communication cost.

Privacy was formally addressed only by P12, which added DP-SGD to FedAvg but dropped accuracy to 79.35%. Personalization was studied in P3, reaching 94.66% using client-level fine-tuning. Vision Transformers entered the FL-DR domain via P4, achieving AUC up to 0.979.

The theoretical foundation for this work is P5, which introduced SCAFFOLD to fix client drift using control variates. However, P5 was never tested on medical imaging — creating a clear oppor-

tunity.

Table 1 summarizes all fifteen papers.

Table 1: Summary of Reviewed Literature on Federated Learning for Diabetic Retinopathy Detection

No.	Year	Paper Title	Objective	Dataset(s)	Model / FL Algorithm	Results	Key Contribution & Limitation
P1	2024	FL for DR Diagnosis: Enhancing Accuracy in Under-Resourced Regions	Build multi-hospital FL for DR keeping data private	DDR, EyePACS, APTOS	EfficientNetB0; FedAvg	Acc: 93.21% (unseen); 91.05% (low-quality)	Multi-hospital FL; mobile/web deployment. <i>Limit: No personalization or DP; FedAvg only.</i>
P2	2025	Privacy Preservation in DR Prediction Using FL (ECSRNet)	Privacy-preserving FL with encryption and reduced cost	Kaggle DR 2015	ShuffleNet+CSPNet+GRU; FedAvg+IKMC+HE	Acc: 98.6%, F1: 98.6%	FedDRNet with encryption and clustering. <i>Limit: Very complex; no SCAFFOLD comparison.</i>
P3	2024	FedDL: Personalized Federated Deep Learning for DR	Build personalized FL model for DR	IDRiD, STARE	Multiple CNNs; FedAvg with personalization	Acc: 94.66%	Client-level personalization in FL for DR. <i>Limit: No DP; no SCAFFOLD; communication cost unstudied.</i>
P4	2023	FL for DR Detection Using Vision Transformers	Use ViT in FL for DR with privacy	APTOS, MESSIDOR-1/2, IDRiD, EyePACS	Vision Transformer (ViT); FedAvg	AUC: 0.910–0.979	First use of ViT in FL for DR. <i>Limit: No personalization; SCAFFOLD not explored; high compute cost.</i>
P5	2020	SCAFFOLD: Stochastic Controlled Averaging for FL	Fix client drift; converge faster on Non-IID data	Simulated, Extended MNIST	Control variates; SCAFFOLD	Faster convergence; fewer rounds vs. FedAvg	Introduced SCAFFOLD; solves client drift. <i>Limit: Not on medical imaging; no built-in privacy.</i>
P6	2022	Federated Transfer Learning for DR Using CNN	Combine transfer learning with FL for DR	EyePACS, Messidor, IDRiD, APTOS	AlexNet (TL); FedAvg and FedProx	CL: 92.19%; FedAvg: 90.07%; FedProx: 85.81%	TL combined with FL; compared FedAvg and FedProx. <i>Limit: No SCAFFOLD; limited Non-IID handling.</i>
P7	2022	Federated Deep Learning for Automated DR Detection	FL-based DR detection with privacy	Kaggle DR	CNN; FedSGD	FedSGD: 97.87%; FL: 96.46%; CNN: 95.02%	First practical FL-DR system. <i>Limit: No Non-IID handling; no advanced FL.</i>
P8	2023	FL for DR in a Multi-center Fundus Screening Network	Test FL in a real multi-center hospital network	OPHDIAT (~697k images)	EfficientNet-B5; FedAvg (weighted/unweighted)	CL AUC: 0.9482; FL AUC: 0.9317–0.9522	Real multi-center FL; FL nearly matches CL. <i>Limit: 15-day training; no advanced FL.</i>
P9	2025	Detection and Classification of DR Using FL	5-class FL-based DR detection with GAN augmentation	Kaggle, EyePACS, Messidor	CNN + GAN; FedAvg (weighted)	Acc: ~97%; Precision: ~100%	FL + GAN + CNN for 5-class grading. <i>Limit: No advanced FL; no personalization; no real deployment.</i>

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(Table 1 continued)

No.	Year	Paper Title	Objective	Dataset(s)	Model / FL Algorithm	Results	Key Contribution & Limitation
P10	2023	DRFL: Federated Learning in DR Grading Using Fundus Images	Federated DR grading with better feature extraction	MESSIDOR-2, IDRiD, Kaggle, hospital	Custom CNN + median CE; FedAvg (modified)	Acc: 98.6%; Spec: 99.3%; F1: 97.5%	DRFL with modified loss function. <i>Limit: No personalization; weak Non-IID handling; no DP.</i>
P11	2025	Combining FL and Transfer Learning with XAI for DR	Unified FL + TL + explainability framework	APTOS, Eye-PACS, Messidor	ResNet + ShuffleNet; Grad-CAM, SHAP; FedAvg+XAI	No experimental results	FL + TL + XAI in one system. <i>Limit: No experiments; no DP; no advanced FL.</i>
P12	2023	Collaborative Differentially Private FL for DR Prediction	DP-based FL for DR with reduced communication cost	Retina dataset (35,120 images, 5 classes)	CNN; FedAvg + DP-SGD	Acc: 83.05% (no DP); 79.35% (with DP); rounds reduced by 49%	Added DP to FL for DR. <i>Limit: Accuracy drops with DP; no SCAFFOLD; no personalization.</i>
P13	2023	FedDDR: A Federated Improved DenseNet for DR Classification	DenseNet-based FL for DR with improved feature reuse	APTOS 2019 (~5,000 images)	DenseNet + MPCE loss; FedAvg	Better accuracy, precision, recall vs. baselines	DenseNet in FL for DR; dense feature reuse. <i>Limit: No DP; no Non-IID handling; no personalization.</i>
P14	2025	DR-FEDPAM: DR Detection Using Federated Proximal Averaging Model	Lightweight FL-based DR detection for resource-limited settings	DDR dataset (13,673 images)	MobileNet; FedProx + GFA (20 clients, 100 rounds)	Acc: 98.36%; beats AlexNet and ResNet	FedProx with lightweight MobileNet and Non-IID. <i>Limit: No XAI; no DP; binary only.</i>
P15	2025	Enhanced DR Classification Using Deep Neural Network	Improve DR classification with DNN over SVM	Kaggle DR	DNN (no FL); Centralized only	Better than SVM-based models	DNN beats classical ML for DR. <i>Limit: No FL; no privacy; not scalable.</i>

. RESEARCH GAP

After reviewing all fifteen papers, six clear gaps were found. Table 2 shows a feature-by-feature comparison.

Gap 1 — No SCAFFOLD for DR Detection: SCAFFOLD corrects client drift and outperforms FedAvg on Non-IID data, yet no reviewed paper has applied it to DR detection or any medical imaging task.

Gap 2 — No IRDRD-Based Federated Study: None of the papers use the Indian Retinal Image Dataset (IRDRD). Models trained on Western datasets may not work well for Indian patients, who make up a large part of the global diabetic population.

Gap 3 — Non-IID Problem Is Largely Ignored: Only P5 and P14 genuinely address Non-IID data. Most other papers assume IID distributions or apply simple weighted averaging without discussing data heterogeneity.

Gap 4 — No Combined SCAFFOLD + Personalization + DP: Each component appears in isolation: personalization in P3, differential privacy in P12, and SCAFFOLD in P5. No paper brings all three together in one system.

Gap 5 — No FedAvg vs. SCAFFOLD Comparison on Retinal Data: No paper directly compares FedAvg and SCAFFOLD on the same retinal dataset under Non-IID conditions, making it impossible to quantify SCAFFOLD's practical benefit for DR.

Gap 6 — Five-Class Grading with Privacy Is Rare: Papers that do five-class grading (P9, P10) do not use advanced FL algorithms or formal privacy mechanisms, limiting their clinical applicability.

Table 2: Gap Analysis: Feature Comparison Across All Reviewed Papers and Proposed System

Feature	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	Prop.
FedAvg	✓	✓	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	✓
FedSGD	—	—	—	—	—	—	✓	—	—	—	—	—	—	—	—	—
FedProx	—	—	—	—	—	✓	—	—	—	—	—	—	—	✓	—	—
SCAFFOLD	—	—	—	—	✓	—	—	—	—	—	—	—	—	—	—	✓
Personalization	—	—	✓	—	—	—	—	—	—	—	—	—	—	—	—	✓
Differential Privacy	—	✓	—	—	—	—	—	—	—	—	—	✓	—	—	—	✓
Non-IID Handling	—	—	—	—	✓	—	—	—	—	—	—	—	—	✓	—	✓
Transfer Learning	—	—	—	—	—	✓	—	—	—	✓	—	—	—	—	—	—
Vision Transformer	—	—	—	✓	—	—	—	—	—	—	—	—	—	—	—	—
GAN Augmentation	—	—	—	—	—	—	—	—	✓	—	—	—	—	—	—	—
Explainable AI (XAI)	—	—	—	—	—	—	—	—	—	—	✓	—	—	—	—	—
DenseNet	—	—	—	—	—	—	—	—	—	—	—	—	✓	—	—	—
MobileNet	—	—	—	—	—	—	—	—	—	—	—	—	—	✓	—	—
Multi-center FL	—	—	—	—	—	—	—	✓	—	✓	—	—	—	—	—	✓
5-Class DR Grading	✓	✓	—	✓	—	—	—	—	✓	✓	—	✓	—	—	✓	✓
IRDRD Dataset	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	✓

. PROPOSED METHODOLOGY

A. Methodology Overview

The proposed system is an FL framework for five-class DR grading. It extends standard FL with: (1) SCAFFOLD aggregation to correct client drift on Non-IID data, (2) local fine-tuning for per-hospital personalization after global convergence, and (3) DP-SGD for formal differential privacy. The local model is a custom CNN. All experiments run on the IRDRD dataset across five simulated federated client nodes.

The system uses Python, PyTorch for deep learning, and Flower (flwr) for federated orchestration. Both FedAvg and SCAFFOLD are compared on identical Non-IID data splits.

B. System Architecture

The central server holds the global CNN and the SCAFFOLD control variates. Each client has a private IRDRD partition and trains locally. Only weight updates are transmitted — no raw patient images ever leave the client. Figure 1 shows the architecture.

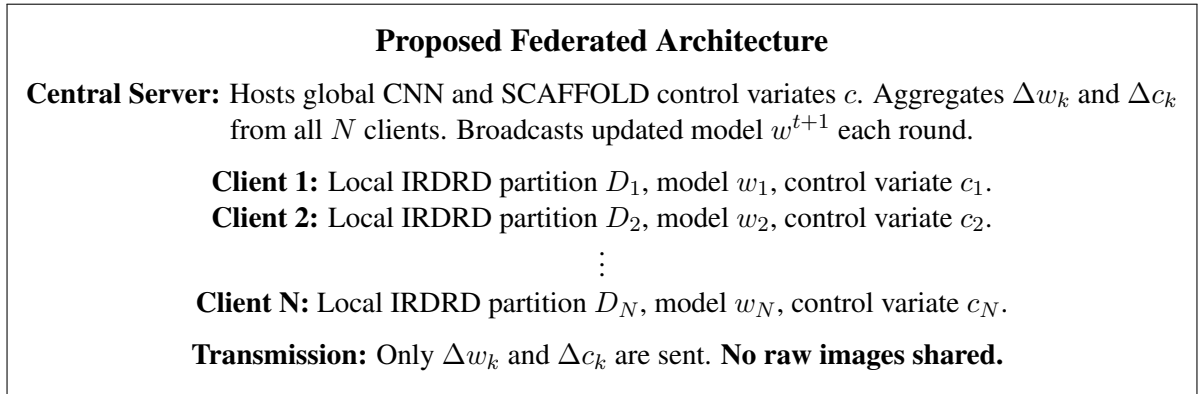


Figure 1: High-level architecture of the proposed federated learning system for DR detection.

C. Step-by-Step Workflow

Step 1. Dataset Partitioning. IRDRD is split into N non-overlapping client partitions using Dirichlet sampling ($\alpha = 0.5$) to simulate Non-IID hospital data. Each partition is split 70% train / 15% validation / 15% test.

Step 2. Local Preprocessing. Each client independently resizes images to 224×224 , applies CLAHE for contrast enhancement, normalizes pixel values, and augments training data (flips, rotation $\pm 15^\circ$, brightness and contrast jitter).

Step 3. Global Model Initialization. The server initializes CNN weights with He initialization. For SCAFFOLD, the global control variate is set to $c^0 = 0$. Both are sent to all N clients.

Step 4. Local Training with DP-SGD. Each client:

- Receives current global model w^t (and c^t for SCAFFOLD).

- Trains for $E = 5$ local epochs using DP-SGD:
 - Clips per-sample gradients to ℓ_2 norm $\leq C = 1.0$.
 - Adds Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ to batch gradients.
- For SCAFFOLD: applies corrected gradient $g_k^{\text{corrected}} = g_k - c_k + c^t$; updates local control variate.
- Computes local update $\Delta w_k = w_k^{t, \text{local}} - w^t$.

Step 5. Update Transmission. Each client sends Δw_k (and Δc_k for SCAFFOLD) to the server. Gaussian noise ensures DP guarantees under the composition theorem.

Step 6. Server Aggregation.

- *FedAvg*: Weighted average of client weights proportional to dataset size.
- *SCAFFOLD*: Weighted average of weight updates applied to global model, plus control variate aggregation to produce c^{t+1} .

Step 7. Broadcast. Updated global model w^{t+1} (and c^{t+1}) is sent to all clients, starting the next round.

Step 8. Personalization. After global convergence, each client freezes the convolutional feature extractor (Blocks 1–4 and GAP) and fine-tunes only the FC layers (FC1, FC2, Output) on local data for 10 epochs using Adam ($\eta = 1 \times 10^{-5}$) and DP-SGD. This adapts the global model to each hospital’s specific data without any further server communication.

D. CNN Model Architecture

Each client uses a custom CNN for five-class DR classification from 224×224 color fundus images.

Layer	Operation	Output Shape	Notes
Input	—	$224 \times 224 \times 3$	RGB retinal image
Block 1	Conv2D (32, 3×3) + BN + ReLU + MaxPool (2×2)	$112 \times 112 \times 32$	Feature extraction
Block 2	Conv2D (64, 3×3) + BN + ReLU + MaxPool (2×2)	$56 \times 56 \times 64$	Feature extraction
Block 3	Conv2D (128, 3×3) + BN + ReLU + MaxPool (2×2)	$28 \times 28 \times 128$	Feature extraction
Block 4	Conv2D (256, 3×3) + BN + ReLU + MaxPool (2×2)	$14 \times 14 \times 256$	Frozen in personalization
GAP	Global Average Pooling	256	Frozen in personalization
FC1	Dense (512, ReLU) + Dropout (0.5)	512	Fine-tuned in personalization
FC2	Dense (256, ReLU) + Dropout (0.3)	256	Fine-tuned in personalization
Output	Dense (5, Softmax)	5	Fine-tuned in personalization

Table 3: Proposed CNN Architecture for DR Classification

Loss: categorical cross-entropy. Optimizer: Adam ($\eta = 1 \times 10^{-4}$). Batch normalization stabilizes training across heterogeneous client data. Global Average Pooling reduces overfitting. Dropout (0.5, 0.3) provides regularization during both federated training and personalization.

E. Federated Learning Algorithms

1) FedAvg (Baseline)

FedAvg is the standard FL baseline [6]. Each round, all N clients train locally for E epochs, then the server aggregates their weights:

$$w^{t+1} = \sum_{k=1}^N \frac{n_k}{n} w_k^{t,\text{local}} \quad (1)$$

where n_k is client k 's dataset size. FedAvg is simple but struggles on Non-IID data because local models drift away from the global optimum during extended local training.

2) SCAFFOLD (Proposed Primary Algorithm)

SCAFFOLD (Stochastic Controlled Averaging for Federated Learning) was introduced by Karimireddy et al. [5] to fix client drift. It assigns a control variate c_k to each client and a global control variate c to the server. These track gradient direction differences and correct local updates so they align with the global optimum.

The corrected local SGD step is:

$$w_k \leftarrow w_k - \eta_l (g_k(w_k) - c_k + c) \quad (2)$$

After local training, client k transmits:

$$\Delta w_k = w_k^{\text{local}} - w^t \quad (3)$$

$$\Delta c_k = \frac{1}{E\eta_l} (w^t - w_k^{\text{local}}) - c + c_k \quad (4)$$

The server updates both model and control variates:

$$w^{t+1} = w^t + \eta_s \cdot \frac{1}{N} \sum_{k=1}^N \Delta w_k \quad (5)$$

$$c^{t+1} = c^t + \frac{1}{N} \sum_{k=1}^N \Delta c_k \quad (6)$$

SCAFFOLD converges in $\mathcal{O}(1/\epsilon^2)$ rounds on Non-IID data, versus $\mathcal{O}(1/\epsilon^{2.5})$ for FedAvg.

3) Differential Privacy via DP-SGD

To go beyond basic FL privacy (not sharing raw data), DP-SGD [12] is applied at each client during local training:

1. Compute per-sample gradients $g_i(w)$.
2. **Clip:** $\tilde{g}_i = g_i / \max(1, \|g_i\|_2/C)$
3. **Add noise:** $\hat{g} = \frac{1}{B}(\sum_i \tilde{g}_i + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}))$
4. **Update weights** using the noisy gradient.

Parameters: $C = 1.0$, $\sigma = 1.1$, $\delta = 10^{-5}$, targeting $\epsilon \leq 5$. DP-SGD is applied during both federated training and personalization.

F. Personalization Strategy

After global convergence, the global CNN is fine-tuned at each client. The feature extractor (Blocks 1–4 + GAP) is frozen; only the FC layers are updated on local data. This combines global feature learning with local adaptation, helping each hospital’s model fit its own patient distribution without extra server communication.

G. FedAvg vs. SCAFFOLD: Comparison Design

Both algorithms run on the same IRDRD Non-IID partitions with identical settings. Table 4 summarizes the experimental design.

Parameter	FedAvg	SCAFFOLD
Dataset	IRDRD (Non-IID)	IRDRD (same partitions)
Clients N	5	5
Dirichlet α	0.5	0.5
Local epochs E	5	5
Communication rounds	100	100
Local LR η_l	1×10^{-4}	1×10^{-4}
Server LR η_s	N/A	1.0
Batch size	32	32
Differential privacy	DP-SGD ($C = 1.0$, $\sigma = 1.1$)	DP-SGD (same)
CNN architecture	Identical	Identical
Evaluation interval	Every 10 rounds	Every 10 rounds
Metrics tracked	Global accuracy, per-class F1, convergence curve	

Table 4: Experimental Comparison Design: FedAvg vs. SCAFFOLD

The hypothesis is that SCAFFOLD will reach 85% accuracy in fewer rounds and achieve higher final accuracy after 100 rounds.

H. Dataset: IRDRD

IRDRD contains high-resolution color fundus photographs annotated by certified ophthalmologists using the five-class ICDR scale. It represents the Indian clinical population.

Dataset usage in the proposed system:

- Partitioned into $N = 5$ non-overlapping client subsets using Dirichlet sampling ($\alpha = 0.5$).
- Each client: 70% train / 15% validation / 15% test (independently).
- Weighted random sampling handles class imbalance (fewer Severe and Proliferative DR cases).
- No images leave any client node at any time.

I. Technologies and Implementation Stack

Component	Tool / Version
Programming Language	Python 3.10
Deep Learning	PyTorch 2.x (CUDA)
Federated Learning	Flower (flwr) – custom SCAFFOLD strategy
Differential Privacy	Opacus (PyTorch DP library)
Image Processing	OpenCV, Albumentations
Data Handling	NumPy, Pandas
Evaluation	Scikit-learn (accuracy, F1, confusion matrix)
Visualization	Matplotlib, Seaborn
Environment	Jupyter Notebook / VS Code
Hardware	NVIDIA GPU (CUDA); multi-core CPU

Table 5: Technology Stack

J. Privacy Mechanism Summary

Privacy is provided at three levels:

1. **FL-level:** Raw images stay at the client. Only model updates are transmitted.
2. **DP-SGD:** Gaussian noise on gradients gives $(\epsilon \leq 5, \delta = 10^{-5})$ -DP, protecting against inference attacks on updates.
3. **Personalization:** DP-SGD is also applied during fine-tuning, so no local patient data leaks from the personalized layers.

. EXPECTED OUTCOMES

The following outcomes are expected after completing experiments on IRDRD:

1. **SCAFFOLD converges faster:** SCAFFOLD is expected to reach 85% accuracy in fewer rounds than FedAvg, based on its proven $\mathcal{O}(1/\epsilon^2)$ convergence guarantee (P5).

2. **Higher final accuracy:** After 100 rounds, SCAFFOLD should reach 88–94% overall accuracy versus 82–88% for FedAvg on the same Non-IID partitions.
3. **Personalization improves local performance:** Each client’s fine-tuned model should outperform the global model on its own test set by 2–5%, especially for clients with skewed class distributions.
4. **DP causes a small accuracy trade-off:** DP-SGD is expected to reduce accuracy by 3–6 percentage points compared to non-private training, consistent with P12 (79.35% with DP vs. 83.05% without).
5. **Five-class grading:** Precision, recall, and F1 will be reported for all five DR grades. Minority classes (Severe, Proliferative DR) will have lower scores due to natural class imbalance, partly offset by weighted sampling.
6. **Privacy-compliant deployment:** No raw images are shared at any stage, satisfying HIPAA and GDPR requirements.

All figures above are projections based on algorithm theory and literature trends. Final quantitative results will be reported after completing experiments.

. CONCLUSION

This paper presented the design of a federated learning system for five-class DR detection that brings together SCAFFOLD, personalization, and differential privacy in a single framework. All experiments run on the IRDRD dataset, which represents Indian patients and has not been used in any prior FL-based DR study.

The literature review confirmed that no existing paper applies SCAFFOLD to medical imaging, and no paper combines all three proposed components. The gap analysis clearly shows this system addresses more research dimensions simultaneously than any prior work.

The FedAvg vs. SCAFFOLD comparison will provide the first empirical evidence of SCAFFOLD’s advantage on retinal data under Non-IID conditions, contributing a practical and reproducible blueprint for privacy-preserving multi-hospital DR screening.

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